

The Salient Features of the Consumer Electronics Linux Forum Specifications

Part 1

From a handheld device that sits comfortably in your palm, to the majestic LCD television that adorns the walls of your house, consumer electronics has become ubiquitous in today's world. Embedded Linux is gaining momentum because of the advances in audio and video technologies, and the user friendly features of Linux. Hence, understanding the significant features of Linux in order to use it in embedded devices is critical for the Linux developer community. Here, we touch upon the salient features of the CELF specifications to expand your knowledge on Linux in the embedded arena.

By 2015, it is estimated that on an average, there will be five consumer electronics devices per person in the developed and developing world. That will add up to more than two trillion devices. If these devices are to be Linux-powered, then what are the important features that Linux must have?

The CELF (Consumer Electronics Linux Forum) community has provided guidelines in the form of CELF specifications, which focus on the significant characteristics of Linux for CE (consumer electronics) devices. Linux has many desirable features that can be used in embedded devices. The top device manufacturers — Sony, Toshiba, NEC, Panasonic, Sharp, Samsung and Philips — have come together to form the CELF and draw up the specifications [1].

The CELF specifications focus on:

- 1) Boot-up time reduction
- 2) Power management
- 3) Audio and video specifications

- 4) Real-time characteristics
- 5) Size reduction
- 6) Linux security

An embedded Linux developer needs to understand how to enable features such as a tickless kernel or device power management, and what can be expected if the feature is in the build. On the other hand, an embedded system architect using Linux will find enough juice in this article to comprehend the broad focus areas, while designing a product powered by Linux. We serve ready-made and easy-to-use information for the embedded system architect, Linux developer and avid reader who want to dig deeper.

While analysing the existing features of the CELF, this review aims to provide a good description of the features, along with information on whether the feature is integrated in the mainline version of the kernel. We will also show you the options to enable the embedded Linux features. Experimental and reported data will be provided to throw more light